Aapa distribution map of Norway

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This script creates a distribution map of appa mires in Norway, for use in reporting for the Bern convention. The resolution is 10x10 km. The distribution is defined based on certain bioclimatic sones and sections.

```
library(tidyverse)
#library(dbConnect)
library(dbplyr)
library(DBI)
library(RPostgres)
library(stars)
library(sf)
library(terra)
library(tmap)
```

Connecting to the database where the bioclimatic sones and sections dataset is

```
# the password is stored in a pgpass.conf file
con <- DBI::dbConnect(drv = RPostgres::Postgres(), host = "t2lippgsql03", dbname = "ano_modul
```

Bring in the data

```
bc <- dplyr::tbl(con, dbplyr::in_schema("helper_variables", "bioclimatic_regions"))

bc <- bc |>
  mutate(geom_wkb = dbplyr::sql("ST_AsBinary(geom)")) |>
  select(-geom) |>
  collect() |>
  mutate(geom = sf::st_as_sfc(geom_wkb, crs = 3035)) |>
  sf::st_as_sf(sf_column_name = "geom")
```

Define distribution

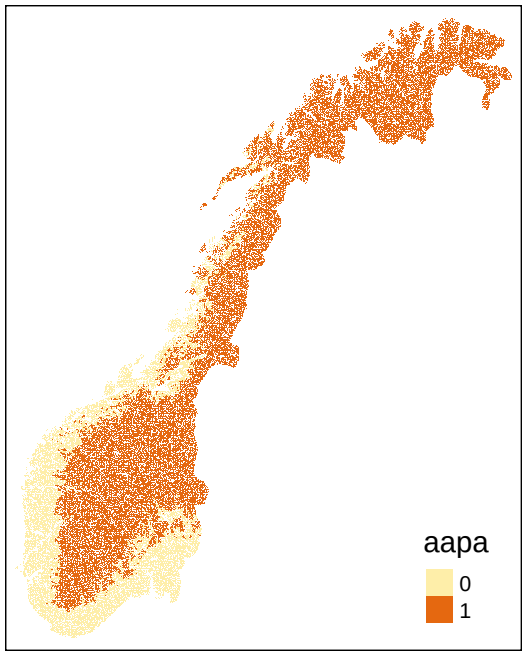
```
reg <- unique(bc$BCregion)
# we exclude boreonemoral and south boreal sones (6SO1-2 and 6SE1)
exl <- grep("6SO-1-2|6SE-1", reg)

# 1 = presence; 0 = absence
bc2 <- bc |>
  mutate(aapa = case_when(
    BCregion %in% reg[exl] ~ 0,
    .default = 1
  ))

# test:
bc2 |> as_tibble() |> count(aapa)
```

```
# A tibble: 2 x 2
   aapa      n
  <dbl>  <int>
1     0  95759
2     1 255643
```

```
# OK
```

A visual test.

```
bc2 |>
  slice_sample(prop = 0.2) |>
  tmap::tm_shape() +
  tmap::tm_fill(col = "aapa", style = "cat")
```



Checking/confirming that the area of the square polygons are 1 km2

```
temp <- bc2 |>
    mutate(km2 = units::drop_units(st_area(geom))*10^-6)

summary(temp |> as_tibble() |> pull(km2))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      1       1       1       1       1       1
```

```
# area is 1 km1
```

```
# defining a summary function
dplyr_summary <- function(data, var) {
  if (inherits(data, "sf")) {
    data <- sf::st_drop_geometry(data)
  }
  data %>%
    summarise(
      Min    = min({{ var }}, na.rm = TRUE),
      Q1     = quantile({{ var }}, 0.25, na.rm = TRUE),
      Median = median({{ var }}, na.rm = TRUE),
```

```
      Mean    = mean({{ var }}, na.rm = TRUE),
      Q3      = quantile({{ var }}, 0.75, na.rm = TRUE),
      Max     = max({{ var }}, na.rm = TRUE)
    )
}
```

Import the EEA 10km standard grid

```
grid <- readRDS(here::here("data", "grid50_kyst_bgr.rds"))

# Test that area is 10 km2
grid |>
  mutate(km2 = units::drop_units(st_area(geometry))*10^-6) |>
  dplyr_summary(km2)
```

```
       Min       Q1  Median     Mean       Q3      Max
1 99.92005 99.98072 100.118 100.1752 100.3278 100.8595
```

```
# the area varies a bit.
```

Creating a regular raster aligned with `grid`.

```
# set resolution
rr <- 10000
# Get bbox
bb <- st_bbox(grid)

# create regular grid frm bbox
gr0 <- rast(xmin= floor(bb["xmin"]/rr)*rr, xmax= ceiling(bb["xmax"]/rr)*rr,
              ymin= floor(bb["ymin"]/rr)*rr, ymax= ceiling(bb["ymax"]/rr)*rr,
              res= rr, crs= "EPSG:3035")
ret <- rasterize(grid, gr0, cover=T, touches=T) # (for very small polygons touches=TRUE is ne

class(ret)
```

```
[1] "SpatRaster"
attr(,"package")
[1] "terra"
```

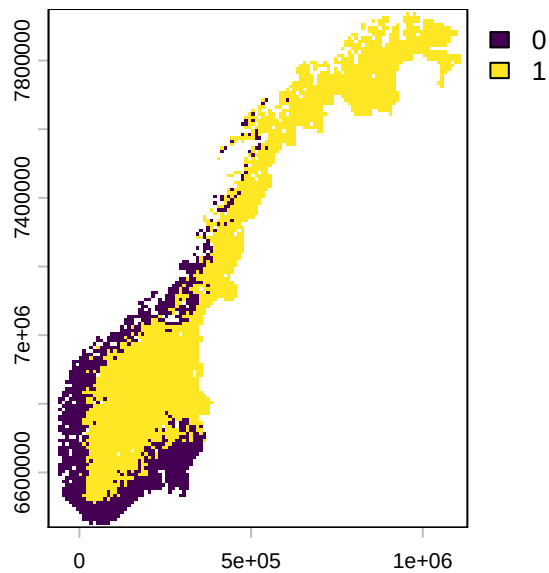
Rasterize and aggregate the 1km2 polygon map

```r
bc_raster_grid <- rasterize(bc2, ret, fun = "max", field = "aapa")
res(bc_raster_grid)
```

```
[1] 10000 10000
```

```r
# OK
```

```r
plot(bc_raster_grid)
```




```r
summary(bc_raster_grid)
```

```
      aapa
 Min.   :0.0000
 1st Qu.:0.0000
 Median :1.0000
 Mean   :0.7239
 3rd Qu.:1.0000
 Max.   :1.0000
 NA's   :14901
```

```r
ncell(bc_raster_grid)
```

```
[1] 18422
```

There are quite a few NA's, but these are from the areas outside mainland Norway (the ocean, Sweden etc.). So that's OK.

Exporting the map

```r
writeRaster(bc_raster_grid, "output/aapa_dist_map.tif", overwrite=TRUE)
```